

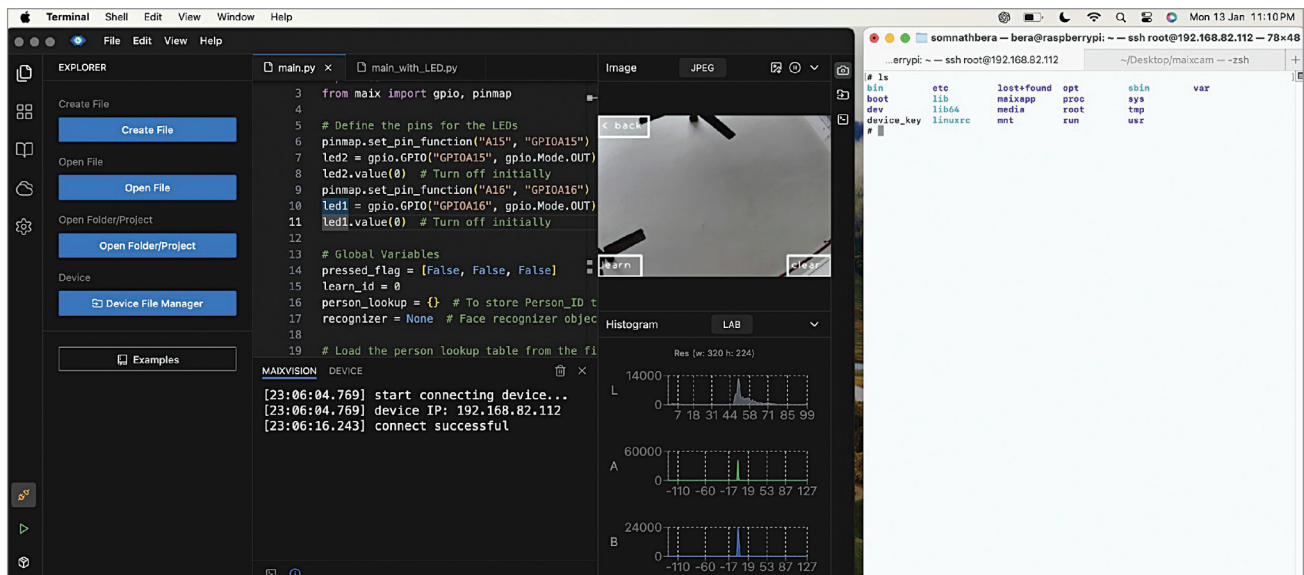


Fig. 7: Device setup for smoker detector

However, note that several of these are in Chinese, and the documentation is often limited. Fig. 7 shows the device setup for the smoker detector.

For uploading

After establishing an SSH connection:

```
ssh root@192.168.82.112
```

(with the password 'root'). Use

SCP to upload the required files—*smoking.mud*, *smoking.cvimodel*, and *smoke_detection_snapshot_metadata.py*—to the board.

For example, to upload *smoking.mud*, use:

```
scp smoking.mud root@192.168.82.112:/root/models/smoking.mud
```

The recommended method is through MaixVision IDE. Within the 'Device File Manager' panel on the right, select the destination directory and press the upload button.

EFY note of caution. Never attempt to upload files to the '/boot' directory. Doing so may corrupt the operating system and require reflash of the image file.

Running code

Running code from the MaixVision IDE is straightforward. First, ensure

that MaixCam is connected to a Wi-Fi network. In the IDE, click the 'Connect' icon in the bottom-left corner to establish a connection with the board. Once connected, press the 'Play' button. The code will then begin execution.

Boot start

This process differs slightly from other Sipeed boards. In MaixVision IDE, click the 'Package' button in the bottom-left corner to create a package of the code. Enter required details such as ID, name, version, developer, and description. Click 'Submit,' then proceed to install the package.

This process creates a link in the '/maixapp/apps/app.info' file with the application name, along with a folder named 'smoker_detection'

inside '/maixapp/apps/', which contains three files, one of which is 'main.py' (the Python code).

Next, open '/maixapp/auto_start.txt' and enter the name of the application: 'smoke_detector'. Save the file. Ensure it contains only this single entry. After disconnecting the MaixCam board, the code will auto-start upon the next boot.

Alternatively, this setup may be completed via the touchscreen interface. Navigate to: Settings→Start Menu→smoker_detection

Restart the board. The 'smoker_detection' application will launch automatically.

Attachment

The system package includes the files 'smoke_detection_snapshot_metadata.py', 'smoking.mud', and 'smoking.cvimodel'. These must be uploaded to the '/root/models/' directory on the MaixCam board. If placed elsewhere, update the file paths in the Python code accordingly. **EFY**

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Somnath Bera is an electronics enthusiast. He is a freelancer and has written several articles across the globe.